

# Supplementary Material (Version B): Environment Sensitivity and Reproducibility of Post-BA SfM Benchmarks

Anonymous 3DV submission

Paper ID 524

## A. What this document supports

This supplementary substantiates four specific statements in the main paper; a reviewer following any of its pointers can go directly to the listed section.

- Protocol, “Track provenance”: “ $\sim 0.209$  is the reproducible level [for the released RESfM checkpoint] ... unchanged (0.217) under the authors’ exact pinned software stack” (Secs. B–D).
- Protocol: “a stack-replication study (supplementary) shows the no-TTT arms are insensitive to it, while test-time fine-tuned means inherit solver-level nondeterminism” (Sec. E).
- Table 2, † footnote: “TTT is run-to-run nondeterministic” (Sec. E).
- Reading any post-BA mean in this literature: differences  $\lesssim 0.05$  are within environment noise (Sec. G).
- Version-B addition — seed robustness of Table 2: every arm and the baseline retrained under five independent seeds; the main claims re-checked pair-by-pair, the stage at which seed variance enters located, and the baseline’s learning-rate sensitivity quantified (Sec. F).

In addition, Sec. H tabulates the per-scene translation, rotation, and registered-camera ( $N_r$ ) results behind the main paper’s dataset-level numbers, with dataset-level rotation and registration summaries in Table 12; Sec. I gives the complete training and evaluation configuration, Sec. J provides the per-scene label-degradation breakdown behind the label-ceiling analysis, and Sec. K provides extended versions of the main paper’s Tables 2, 5, 6, and 7 (adding rotation and  $N_r$ ) intended to replace them in the camera-ready.

## B. The reproduction question

Evaluating the RESfM authors’ released checkpoint in our harness on the 36 MegaDepth test scenes yields a post-BA mean translation error of 0.209 vs. the published 0.169 (rotation  $1.89^\circ$  vs.  $1.29^\circ$ ). Since our

controlled comparisons hinge on baseline fidelity, we traced this discrepancy to its source. The finding is of independent interest: post-BA means on internet-photo benchmarks carry a practical environment-noise floor of  $\mathcal{O}(0.05)$ , produced by chaotic basin selection in robust bundle adjustment.

Table 8. Released RESfM checkpoint, MegaDepth 36-scene post-BA means, under three conditions, vs. the published number.  $N_r$  is the mean number of registered cameras per scene; the published row averages the paper’s per-scene Table 1 values.

| Condition                           | Trans | Rot ( $^\circ$ ) | $N_r$ |
|-------------------------------------|-------|------------------|-------|
| Published (paper Table 1)           | 0.169 | 1.29             | 298   |
| Our env (pip/OpenBLAS), 5-seed mean | 0.203 | 1.82             | 304   |
| Our env, seed 20                    | 0.209 | 1.89             | 304   |
| Authors’ exact stack (conda/MKL)    | 0.217 | 1.92             | 304   |

## C. Eliminating the controllable factors

Data. Our MegaDepth npz files are byte-identical to the authors’ released archive, including the Group-2 300-camera subsamples.

Randomness. Five evaluation seeds vary the pre-BA network output by  $\pm 0.004$  and the post-BA mean by  $\pm 0.01$ ; negligible.

Network forward pass. The pre-BA output (translation/rotation  $\sim 1.73/11.9^\circ$ ) is essentially seed- and environment-stable; the divergence therefore enters at the robust-BA stage. (Notably, the paper does not report pre-BA numbers; all its tables are post-robust-BA.)

Software stack. We rebuilt the authors’ exact pinned environment from their released environment.yaml: Python 3.8.20, MKL-backed numpy 1.24.3 (mkl=2023.1.0), pycolmap 3.10.0, pyceres 2.3, lightning 2.3.3. (The sole deviation is the CUDA build of PyTorch, forced by GPU-driver compatibility; the BA stage is CPU-only.) Re-evaluating all 36 scenes in this

matched environment gives translation/rotation errors of  $0.217/1.92^\circ$ , statistically indistinguishable from our environment's  $0.209/1.89^\circ$ , and not the published  $0.169/1.29^\circ$ . Registration is equally stable: all three of our conditions register 304 cameras per scene on average (Table 8), slightly more than the published 298, so the higher mean error is not an artifact of evaluating fewer cameras.

## D. Mechanism: chaotic basin selection in robust BA

Initialization perturbation is at the  $10^{-10}$  level. We isolated the BLAS layer by running the identical DLT triangulation (the BA initialization) on identical inputs under OpenBLAS and MKL, single-threaded: over 11,131 triangulated points the maximum relative difference is  $4.8 \times 10^{-10}$  (mean  $5 \times 10^{-14}$ ; NaN patterns identical).

Yet outcomes move by  $\mathcal{O}(0.1)$ . This sub-nanoscale perturbation changes the per-scene post-BA translation by up to 0.167, with 16/36 scenes moving by more than 0.01. Robust BA on ill-conditioned scenes amplifies last-bit differences into different convergence basins.

The published gap has the same signature. Comparing our released-checkpoint evaluation to the paper's Table 1 per scene: 20/36 scenes agree within 0.05 (8 essentially exactly), 17 are better in our runs, and the entire mean gap is carried by four basin flips (scenes 0046, 0148, 0012, 5016; e.g. 0046:  $0.043 \rightarrow 0.892$ ). Bidirectional, scene-concentrated flips are the signature of basin selection, not of a weaker checkpoint, which would shift errors in one direction rather than improve 17 of 36 scenes.

## E. Stack replication and TTT run-to-run variance

Beyond the released checkpoint, we re-evaluated both from-scratch main-comparison checkpoints under the authors' pinned stack (MKL numpy, pycolmap 3.10.0, pyceres 2.3), changing nothing else, and additionally repeated the TTT evaluation within our own stack. The supervised baseline is essentially unchanged across stacks: MegaDepth  $0.344 \rightarrow 0.350$  mean,  $0.070 \rightarrow 0.088$  median, only 7/36 scenes moving more than 0.05 (max  $\pm 0.37$ , bidirectional). This matches the released-checkpoint result above. The SS-RESfM weight+TTT arm, by contrast, varies regardless of stack: three evaluations of the same checkpoint (ours, a same-stack repeat, the authors' stack) give MegaDepth means 0.316, 0.401, 0.453 ( $\sim 18$ –20/36 scenes shifting per repeat, up to  $\pm 2.9$ , largely the same scenes, medians 0.12–0.19). The attribution is therefore not the software stack but

test-time fine-tuning itself: it runs an optimization at evaluation time, so the chaotic basin selection documented above re-enters the result on every run. Consequently the paper reports all results from the single declared pipeline, treats TTT means as carrying  $\sim \pm 0.07$  run-to-run variance, and rests its environment- and run-independent claims on the deterministic no-TTT cells (which reproduce exactly on repeat).

## F. Training-seed variability: five-seed bands

After the submission freeze we retrained every arm and the supervised baseline under five independent training seeds (20–24; five networks per recipe, identical configuration otherwise) and ran the full evaluation for each. Table 9 reports, per cell, the five-seed band of the network output before bundle adjustment (pre-BA, after per-scene fine-tuning) and after robust BA (post-BA, the paper's headline stage). Main-paper Table 2 reports the seed-20 column of this program.

Eval-seed control. Re-evaluating the authors' fixed released weights under five evaluation seeds gives  $0.203 \pm 0.008$  on MegaDepth,  $10.71 \pm 0.27$  on 1DSfM, and values identical to two decimals on Strecha and BlendedMVS: with the network held constant, the evaluation is close to deterministic. The bands in Table 9 therefore measure genuine differences between independently trained networks, not evaluation noise.

The paper's claims under seed pairing. The high-contamination results are seed-robust in the strongest sense: at 60% contamination the worst plain-removal seed (15.8) stays below the best baseline seed (16.2), and at 43% every removal median (1.4–1.9) beats every baseline median (2.4–4.3). The Olsson clean-benchmark collapse of the supervised baseline recurs in four of five seeds (the fifth lands at 2.69 — basin-dependent escape), and also in the released checkpoint (9.10) and an authors-faithful retrain (8.76), while every seed of every label-free arm lands at 2.9–3.6 (weight+TTT excepted). On BlendedMVS the mad/weight arms ( $0.14 \pm 0.02$ ) sit below every baseline seed ( $0.35 \pm 0.04$ ). Conversely, the baseline's indistribution MegaDepth advantage persists at band level ( $0.37 \pm 0.12$  vs.  $0.51 \pm 0.08$  for madweight) but is itself seed-volatile (individual baseline seeds span 0.19–0.53).

Where the variance enters. On the contaminated datasets pre-BA bands are tight for all methods ( $\pm 0.02$ – $0.13$  for the label-free arms,  $\pm 0.05$ – $0.43$  for the baseline) while the same cells after BA carry  $\pm 1.7$ – $4.1$ : BA lowers the error almost everywhere (234 of 242 baseline scene evaluations examined) yet widens the relative seed band by an order of magnitude — the seed-level counterpart of the environment-level basin

selection of Sec. D. The one exception is the baseline’s clean-OOD cells, whose variance predates BA (Strecha pre-BA  $0.68 \pm 0.78$ , Olsson  $11.01 \pm 2.27$ ) and is bimodal per scene: entry-P10 costs 0.04–0.06 on three seeds and 4.0/7.0 on the other two; dtu106 sits at 250–260 on four seeds and 53 on one. The label-free arms never show this mode (per-scene stds down to 0.01): worse pre-BA on clean data, but reproducible under retraining — consistent with a supervised head, fit to labels generated at  $\sim 25\%$  contamination, being underdetermined on near-outlier-free scenes.

Learning-rate sensitivity. Retraining the baseline at  $10^{-4}$  (vs. RESfM’s released  $10^{-3}$ , which the paper declares) improves its contaminated-OOD results (1DSfM 5.90, 1DSfM-hard 12.81, BlendedMVS 0.14) at the cost of in-distribution and Strecha (0.53 and 1.73); a 30k-epoch replicate plateaus mid-schedule, so this is a regularization effect, not under-training. The Olsson collapse persists at both learning rates ( $\sim 8.3$ – $8.5$ ), and the symmetric change on our side ( $10^{-3}$  for the label-free arms) helps nowhere — so the per-method learning-rate policy is not tuned in our favor, and the clean-benchmark finding is lr-robust while the contaminated-OOD margins narrow against a  $10^{-4}$  baseline.

## G. Implications for reading the paper’s numbers

(1) The published number and ours are draws from a distribution over environments; no software matching can force a particular draw. (2) Differences of  $\lesssim 0.05$  in post-BA translation means on these benchmarks are within environment noise and should not be read as method differences, by any paper. (3) Consequently, meaningful comparisons must be environment-controlled: every number in the main paper (all arms, RESfM-from-scratch, and the released checkpoint) is produced in a single environment, and the out-of-distribution margins ( $2\times$  and larger) sit far above this noise floor. (4) With the five-seed program of Sec. F, the paper’s single-seed caveat is retired: the high-contamination and clean-benchmark findings hold under every seed pairing, and the remaining seed sensitivity is confined to the in-distribution margin and the baseline’s clean-OOD cells.

## H. Per-scene results

Tables 10 and 11 report the per-scene post-BA translation and rotation errors and registered-camera counts underlying the main paper’s dataset-level numbers (seed 20, the declared pipeline): the 36 MegaDepth test scenes and the ten 1DSfM scenes, for the three self-

supervised arms (madweight, weight, weight+TTT) and the supervised RESfM baseline trained from scratch under the identical recipe. The per-scene view makes two aggregate statements concrete: on 1DSfM no single arm dominates, the best mechanism flips scene by scene (with *Ellis\_Island* and *Tower\_of\_London* hard for every arm), consistent with the per-scene complementarity analysis in the main paper; on MegaDepth the four arms are close on most scenes and the dataset-level gap is dominated by a handful of high-error scenes. Per scene, the catastrophic scenes (*Ellis\_Island*, *Tower\_of\_London*) fail in both metrics, though rotation does not always track translation (weight’s *Vienna\_Cathedral* failure is translation-specific: 22.71/0.46), and  $N_r$  stays in the 58–92% range of each scene’s images on 1DSfM with no arm systematically dropping scenes. Table 12 completes the dataset-level record with rotation and registered-camera counts alongside translation. Rotation broadly agrees with translation at the dataset level: the supervised BlendedMVS failure is even larger in rotation ( $36^\circ$ ) and the high-contamination ordering is preserved, though the best arm can differ per metric (on MegaDepth, weight+TTT leads translation while RESfM leads rotation). Registration is contamination-dependent: the supervised baseline registers the most cameras in distribution (85% vs. 78–80% on MegaDepth) and on clean Strecha, while under high contamination the label-free arms retain more cameras than supervision (38–43% vs. 37% on 1DSfM-hard) at lower error (madweight).

## I. Implementation details

Table 13 lists the full training and evaluation configuration. Both methods share every setting except the base learning rate ( $10^{-4}$  self-supervised,  $10^{-3}$  supervised, each selected on its own training objective) and the supervision source (COLMAP-derived labels vs. the adaptive percentile pseudo-labels).

Table 9. Pre-BA  $\rightarrow$  post-BA translation error, mean $\pm$ std over five training seeds (20–24); evaluation configuration fixed. Pre-BA bands are one to two orders of magnitude tighter than post-BA bands, except the supervised baseline’s clean-OOD cells (Strecha, Olsson), whose variance predates BA (see text). <sup>§</sup>For the released checkpoint the MegaDepth entry is a band over five evaluation seeds on the fixed weights: re-seeding the evaluation moves the result by two orders of magnitude less than retraining does.

| Arm                                                                   | MegaDepth                                         | 1DSM                                            | 1DSM-hard                                       | Strecha                                       | BlendedMVS                                    | Olsson                                         |
|-----------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------|-------------------------------------------------|-----------------------------------------------|-----------------------------------------------|------------------------------------------------|
| madweight                                                             | 3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08     | 26.14 $\pm$ 0.04 $\rightarrow$ 11.04 $\pm$ 1.68 | 36.33 $\pm$ 0.07 $\rightarrow$ 18.92 $\pm$ 4.05 | 3.77 $\pm$ 0.24 $\rightarrow$ 2.65 $\pm$ 0.25 | 0.58 $\pm$ 0.03 $\rightarrow$ 0.14 $\pm$ 0.02 | 16.38 $\pm$ 0.45 $\rightarrow$ 3.01 $\pm$ 0.34 |
| weight                                                                | 3.04 $\pm$ 0.03 $\rightarrow$ 0.51 $\pm$ 0.09     | 26.60 $\pm$ 0.09 $\rightarrow$ 14.35 $\pm$ 2.97 | 36.26 $\pm$ 0.04 $\rightarrow$ 22.92 $\pm$ 2.79 | 2.57 $\pm$ 0.19 $\rightarrow$ 1.97 $\pm$ 0.09 | 0.51 $\pm$ 0.01 $\rightarrow$ 0.14 $\pm$ 0.03 | 15.86 $\pm$ 0.28 $\rightarrow$ 2.91 $\pm$ 0.33 |
| weight+TTT                                                            | 3.01 $\pm$ 0.03 $\rightarrow$ 0.40 $\pm$ 0.11     | 26.63 $\pm$ 0.10 $\rightarrow$ 15.72 $\pm$ 2.10 | 36.20 $\pm$ 0.12 $\rightarrow$ 22.24 $\pm$ 3.17 | 2.63 $\pm$ 0.21 $\rightarrow$ 2.00 $\pm$ 0.05 | 0.50 $\pm$ 0.02 $\rightarrow$ 0.14 $\pm$ 0.02 | 13.41 $\pm$ 0.19 $\rightarrow$ 8.03 $\pm$ 0.30 |
| remove                                                                | 3.03 $\pm$ 0.02 $\rightarrow$ 0.51 $\pm$ 0.08     | 25.34 $\pm$ 0.10 $\rightarrow$ 9.13 $\pm$ 2.17  | 36.32 $\pm$ 0.09 $\rightarrow$ 12.42 $\pm$ 2.05 | 4.22 $\pm$ 0.19 $\rightarrow$ 3.20 $\pm$ 0.11 | 0.62 $\pm$ 0.01 $\rightarrow$ 0.24 $\pm$ 0.06 | 17.05 $\pm$ 0.39 $\rightarrow$ 3.41 $\pm$ 0.57 |
| remove+TTT                                                            | 3.04 $\pm$ 0.02 $\rightarrow$ 0.56 $\pm$ 0.07     | 25.23 $\pm$ 0.13 $\rightarrow$ 8.93 $\pm$ 1.20  | 36.32 $\pm$ 0.07 $\rightarrow$ 15.15 $\pm$ 0.98 | 4.31 $\pm$ 0.18 $\rightarrow$ 3.01 $\pm$ 0.07 | 0.63 $\pm$ 0.02 $\rightarrow$ 0.22 $\pm$ 0.10 | 17.03 $\pm$ 0.47 $\rightarrow$ 3.63 $\pm$ 0.35 |
| RESfm-scratch (supervised)                                            | 1.76 $\pm$ 0.05 $\rightarrow$ 0.37 $\pm$ 0.12     | 22.13 $\pm$ 0.27 $\rightarrow$ 10.50 $\pm$ 1.45 | 35.80 $\pm$ 0.43 $\rightarrow$ 19.16 $\pm$ 3.42 | 0.68 $\pm$ 0.78 $\rightarrow$ 0.39 $\pm$ 0.72 | 0.45 $\pm$ 0.01 $\rightarrow$ 0.35 $\pm$ 0.04 | 11.01 $\pm$ 2.27 $\rightarrow$ 7.37 $\pm$ 2.63 |
| single-run reference rows (no seed band; pre $\rightarrow$ post only) |                                                   |                                                 |                                                 |                                               |                                               |                                                |
| RESfm-faithful                                                        | 1.94 $\rightarrow$ 0.50                           | 22.26 $\rightarrow$ 9.75                        | 35.41 $\rightarrow$ 17.58                       | 0.29 $\rightarrow$ 0.14                       | 0.44 $\rightarrow$ 0.37                       | 12.01 $\rightarrow$ 8.76                       |
| RESfm-faithful @10 <sup>-4</sup>                                      | 2.23 $\rightarrow$ 0.60                           | 22.88 $\rightarrow$ 6.72                        | 34.98 $\rightarrow$ 16.21                       | 1.27 $\rightarrow$ 0.22                       | 0.32 $\rightarrow$ 0.17                       | 12.00 $\rightarrow$ 8.11                       |
| RESfm-released <sup>§</sup>                                           | 1.730 $\pm$ 0.005 $\rightarrow$ 0.203 $\pm$ 0.008 | 22.42 $\rightarrow$ 10.71 $\pm$ 0.27            | 35.26 $\rightarrow$ 15.29                       | 0.97 $\rightarrow$ 0.20 $\pm$ 0.00            | 0.42 $\rightarrow$ 0.33 $\pm$ 0.00            | 12.11 $\rightarrow$ 9.10                       |
| RESfm @10 <sup>-4</sup> , 30k                                         | 2.08 $\rightarrow$ 0.36                           | 23.14 $\rightarrow$ 6.12                        | 35.76 $\rightarrow$ 13.88                       | 1.92 $\rightarrow$ 1.64                       | 0.29 $\rightarrow$ 0.18                       | 11.94 $\rightarrow$ 8.27                       |

Table 10. Per-scene MegaDepth results (layout follows RESfm’s Table 1). For each scene we show the number of input images ( $N_c$ ) and the fraction of outlier observations; for each model, the number of registered cameras ( $N_r$ ) and mean rotation ( $e_R$ , degrees) and translation ( $e_t$ ) errors, post-BA, seed 20. Above the middle rule are Group 1 scenes ( $< 1000$  images); below are Group 2 scenes ( $> 1000$  images, subsampled to 300 for testing), each group ordered by outlier fraction. Winning results per scene are marked in bold and underlined. weight+TTT values are single draws of a run-to-run nondeterministic evaluation (Sec. E).

| Scene | $N_c$ | %out | madweight |                    |                    | weight |                    |                    | weight+TTT |                    |                    | RESfm |                    |                    |
|-------|-------|------|-----------|--------------------|--------------------|--------|--------------------|--------------------|------------|--------------------|--------------------|-------|--------------------|--------------------|
|       |       |      | $N_r$     | $e_R$              | $e_t$              | $N_r$  | $e_R$              | $e_t$              | $N_r$      | $e_R$              | $e_t$              | $N_r$ | $e_R$              | $e_t$              |
| 0238  | 522   | 45%  | 437       | 19.02              | 1.41               | 137    | 8.23               | 0.62               | 141        | 8.37               | 0.63               | 355   | <b><u>1.40</u></b> | <b><u>0.12</u></b> |
| 0060  | 528   | 42%  | 469       | 0.29               | 0.03               | 476    | 0.20               | 0.02               | 452        | 4.13               | 0.37               | 494   | <b><u>0.04</u></b> | <b><u>0.01</u></b> |
| 0197  | 870   | 41%  | 470       | 7.22               | 0.89               | 341    | 3.54               | 0.38               | 422        | 6.94               | 1.11               | 739   | <b><u>0.97</u></b> | <b><u>0.11</u></b> |
| 0094  | 763   | 40%  | 493       | 2.72               | 0.45               | 505    | 1.44               | <b><u>0.16</u></b> | 432        | <b><u>1.34</u></b> | 0.17               | 572   | 5.33               | 0.88               |
| 0265  | 571   | 39%  | 457       | 17.62              | 3.26               | 477    | 5.78               | 2.34               | 547        | <b><u>0.60</u></b> | <b><u>0.19</u></b> | 424   | 16.27              | 3.09               |
| 0083  | 635   | 31%  | 342       | 4.34               | 0.23               | 511    | 17.70              | 1.56               | 317        | 3.20               | 0.20               | 582   | <b><u>1.40</u></b> | <b><u>0.10</u></b> |
| 0076  | 558   | 30%  | 423       | 1.40               | 0.38               | 519    | 0.61               | 0.23               | 454        | 0.74               | 0.22               | 499   | <b><u>0.25</u></b> | <b><u>0.02</u></b> |
| 0185  | 368   | 30%  | 347       | 0.21               | 0.01               | 335    | 0.24               | 0.04               | 321        | 0.88               | 0.09               | 352   | <b><u>0.07</u></b> | <b><u>0.01</u></b> |
| 0048  | 512   | 24%  | 475       | 1.93               | 0.14               | 467    | <b><u>0.35</u></b> | <b><u>0.02</u></b> | 450        | 0.43               | 0.03               | 489   | 1.85               | 0.11               |
| 0024  | 356   | 23%  | 299       | 1.27               | 0.18               | 291    | <b><u>1.13</u></b> | <b><u>0.17</u></b> | 289        | 4.78               | 1.13               | 334   | 3.67               | 1.09               |
| 0223  | 214   | 17%  | 205       | 2.66               | 0.24               | 198    | 1.69               | 0.23               | 199        | 3.06               | 0.30               | 204   | <b><u>0.45</u></b> | <b><u>0.08</u></b> |
| 5016  | 28    | 17%  | 23        | 0.27               | 0.04               | 25     | 0.08               | 0.02               | 28         | <b><u>0.07</u></b> | 0.02               | 28    | 0.08               | <b><u>0.01</u></b> |
| 0046  | 440   | 15%  | 422       | <b><u>2.88</u></b> | <b><u>0.22</u></b> | 421    | 4.11               | 0.30               | 419        | 3.88               | 0.25               | 410   | 14.64              | 0.93               |
| 0099  | 299   | 47%  | 224       | <b><u>0.81</u></b> | 0.15               | 216    | 2.77               | 3.52               | 212        | 1.58               | <b><u>0.13</u></b> | 252   | 2.18               | 0.39               |
| 1001  | 285   | 44%  | 280       | 1.58               | 1.22               | 278    | 1.49               | 0.73               | 279        | <b><u>1.22</u></b> | <b><u>0.49</u></b> | 273   | 1.23               | 0.50               |
| 0231  | 296   | 42%  | 274       | 1.63               | 0.15               | 257    | 0.64               | 0.07               | 264        | 1.49               | 0.24               | 265   | <b><u>0.17</u></b> | <b><u>0.02</u></b> |
| 0411  | 299   | 30%  | 281       | 0.22               | 0.03               | 245    | 2.99               | 0.32               | 250        | 4.12               | 0.46               | 275   | <b><u>0.11</u></b> | <b><u>0.02</u></b> |
| 0377  | 295   | 28%  | 200       | 0.62               | 0.05               | 222    | 2.76               | 0.26               | 243        | <b><u>0.56</u></b> | <b><u>0.03</u></b> | 223   | 0.82               | 0.06               |
| 0102  | 299   | 26%  | 289       | 0.15               | 0.02               | 289    | <b><u>0.07</u></b> | <b><u>0.01</u></b> | 291        | 0.08               | 0.01               | 287   | 0.10               | 0.02               |
| 0147  | 298   | 25%  | 261       | 31.69              | 2.99               | 235    | 15.28              | 1.21               | 225        | 15.27              | <b><u>1.20</u></b> | 247   | <b><u>4.61</u></b> | 3.22               |
| 0148  | 287   | 25%  | 220       | 2.60               | 0.33               | 201    | 2.11               | 0.11               | 196        | <b><u>0.62</u></b> | 0.04               | 202   | 0.94               | <b><u>0.04</u></b> |
| 0446  | 298   | 22%  | 292       | <b><u>0.40</u></b> | <b><u>0.03</u></b> | 285    | 1.20               | 0.19               | 292        | 1.04               | 0.19               | 293   | 0.71               | 0.05               |
| 0022  | 297   | 21%  | 275       | 0.14               | 0.02               | 272    | 0.22               | 0.03               | 267        | 0.33               | 0.04               | 276   | <b><u>0.11</u></b> | <b><u>0.02</u></b> |
| 0327  | 298   | 21%  | 282       | 1.97               | 0.09               | 285    | 0.22               | 0.02               | 284        | 0.21               | 0.02               | 277   | <b><u>0.09</u></b> | <b><u>0.01</u></b> |
| 0015  | 284   | 21%  | 196       | 2.64               | 0.30               | 196    | 3.87               | 0.56               | 211        | 3.90               | 0.65               | 232   | <b><u>1.31</u></b> | <b><u>0.30</u></b> |
| 0455  | 298   | 20%  | 283       | <b><u>0.20</u></b> | <b><u>0.01</u></b> | 285    | 0.21               | 0.01               | 288        | 0.46               | 1.60               | 293   | 0.85               | 0.09               |
| 0496  | 297   | 19%  | 279       | <b><u>0.27</u></b> | 0.04               | 285    | 0.50               | <b><u>0.04</u></b> | 287        | 1.28               | 0.09               | 283   | 0.46               | 0.07               |
| 1589  | 299   | 17%  | 265       | 0.96               | 0.06               | 262    | 0.28               | 0.05               | 296        | 1.91               | 0.30               | 291   | <b><u>0.06</u></b> | <b><u>0.01</u></b> |
| 0012  | 299   | 16%  | 281       | <b><u>0.23</u></b> | <b><u>0.02</u></b> | 247    | 9.51               | 0.70               | 262        | 2.76               | 0.22               | 288   | 5.53               | 0.34               |
| 0104  | 284   | 16%  | 189       | 0.32               | 0.03               | 193    | 2.28               | 0.18               | 194        | 1.44               | 0.07               | 199   | <b><u>0.30</u></b> | <b><u>0.02</u></b> |
| 0019  | 299   | 15%  | 232       | 3.22               | 0.17               | 253    | 6.55               | 0.28               | 264        | 3.24               | 0.13               | 252   | <b><u>0.05</u></b> | <b><u>0.01</u></b> |
| 0063  | 293   | 14%  | 258       | 0.63               | 0.08               | 268    | 0.72               | 0.07               | 267        | 0.65               | <b><u>0.06</u></b> | 266   | <b><u>0.44</u></b> | 0.07               |
| 0130  | 285   | 14%  | 250       | 10.17              | 1.60               | 184    | 1.87               | 0.16               | 185        | <b><u>0.33</u></b> | <b><u>0.01</u></b> | 195   | 1.03               | 0.07               |
| 0080  | 284   | 13%  | 132       | 1.10               | <b><u>0.05</u></b> | 148    | 2.48               | 0.20               | 147        | 0.70               | 0.07               | 141   | <b><u>0.66</u></b> | 0.10               |
| 0240  | 298   | 12%  | 275       | <b><u>2.23</u></b> | <b><u>0.25</u></b> | 275    | 4.64               | 0.34               | 272        | 5.32               | 0.46               | 282   | 3.02               | 0.35               |
| 0007  | 290   | 12%  | 106       | <b><u>0.07</u></b> | <b><u>0.01</u></b> | 269    | 1.86               | 0.09               | 261        | 3.28               | 0.14               | 104   | 0.99               | 0.04               |

Table 11. Per-scene 1DSfM results (layout follows RESfM’s Table 1). Columns as in Table 10; seed 20, scenes ordered by outlier fraction. Winning results per scene are marked in bold and underlined.

| Scene               | $N_c$ | %out | madweight |                     |                    | weight |                    |                     | weight+TTT |                     |                    | RESfM |                    |                    |
|---------------------|-------|------|-----------|---------------------|--------------------|--------|--------------------|---------------------|------------|---------------------|--------------------|-------|--------------------|--------------------|
|                     |       |      | $N_r$     | $e_R$               | $e_t$              | $N_r$  | $e_R$              | $e_t$               | $N_r$      | $e_R$               | $e_t$              | $N_r$ | $e_R$              | $e_t$              |
| Ellis_Island        | 227   | 56%  | 180       | <b><u>17.16</u></b> | 17.59              | 171    | 17.37              | <b><u>15.90</u></b> | 177        | 17.74               | 17.11              | 192   | 18.15              | 16.02              |
| NYC_Library         | 332   | 54%  | 282       | 2.32                | 2.94               | 261    | <b><u>1.16</u></b> | <b><u>1.19</u></b>  | 279        | 10.30               | 3.39               | 285   | 12.99              | 6.28               |
| Vienna_Cathedral    | 836   | 51%  | 537       | 7.42                | 7.30               | 484    | <b><u>0.46</u></b> | 22.71               | 651        | 26.80               | 33.57              | 492   | 1.09               | <b><u>2.29</u></b> |
| Notre_Dame          | 553   | 48%  | 507       | 3.33                | 0.68               | 491    | 1.00               | 0.34                | 485        | <b><u>0.83</u></b>  | <b><u>0.33</u></b> | 493   | 1.08               | 0.49               |
| Yorkminster         | 437   | 45%  | 299       | 8.05                | <b><u>5.79</u></b> | 371    | 14.95              | 30.87               | 380        | 12.20               | 10.79              | 309   | <b><u>6.88</u></b> | 8.77               |
| Piazza_del_Popolo   | 338   | 42%  | 218       | <b><u>0.43</u></b>  | <b><u>0.94</u></b> | 265    | 19.54              | 10.66               | 254        | 20.46               | 17.05              | 255   | 3.15               | 2.49               |
| Tower_of_London     | 472   | 42%  | 317       | 35.61               | 76.34              | 337    | 23.30              | <b><u>28.29</u></b> | 314        | <b><u>17.91</u></b> | 72.61              | 299   | 49.66              | 67.95              |
| Montreal_Notre_Dame | 450   | 36%  | 311       | 0.59                | 0.94               | 325    | <b><u>0.55</u></b> | 11.35               | 320        | 0.81                | 1.24               | 400   | 0.59               | <b><u>0.93</u></b> |
| Alamo               | 577   | 29%  | 467       | 2.45                | 1.77               | 507    | 2.74               | 2.18                | 497        | 2.49                | <b><u>1.67</u></b> | 502   | <b><u>2.42</u></b> | 1.73               |
| Madrid_Metropolis   | 341   | 29%  | 249       | 5.72                | 9.63               | 301    | 2.84               | 4.37                | 295        | 2.35                | 3.52               | 287   | <b><u>1.82</u></b> | <b><u>3.38</u></b> |

Table 12. Dataset-level translation, rotation, and registered cameras (post-BA, seed 20) for the four main-comparison arms. Per-scene values are camera means;  $N_r$  is the fraction of ground-truth cameras surviving robust BA and the 5-pixel reprojection quality mask, summed over the dataset’s scenes.

|                                                        | madweight   | weight      | weight+TTT  | RESfM       |
|--------------------------------------------------------|-------------|-------------|-------------|-------------|
| Translation error (mean / median over scenes)          |             |             |             |             |
| MegaDepth                                              | 0.422/0.145 | 0.423/0.182 | 0.316/0.186 | 0.344/0.07  |
| 1DSfM                                                  | 12.4/4.37   | 12.8/11.0   | 16.1/7.16   | 11.0/2.93   |
| 1DSfM-hard                                             | 12.2/8.83   | 21.9/13.1   | 26.4/17.7   | 16.2/13.1   |
| Strecha                                                | 2.5/0.97    | 2/0.026     | 1.98/0.005  | 0.006/0.005 |
| BlendedMVS                                             | 0.146/0.004 | 0.1/0.036   | 0.136/0.011 | 0.379/0.348 |
| Olsson                                                 | 2.57/1.15   | 2.65/1.22   | 7.74/0.18   | 8.47/0.17   |
| Rotation error ( $^\circ$ , mean / median over scenes) |             |             |             |             |
| MegaDepth                                              | 3.49/1.33   | 3.04/1.77   | 2.51/1.39   | 2.01/0.84   |
| 1DSfM                                                  | 8.31/4.53   | 8.39/2.79   | 11.19/11.25 | 9.79/2.79   |
| 1DSfM-hard                                             | 14.40/7.39  | 25.58/17.70 | 31.97/12.83 | 22.70/15.86 |
| Strecha                                                | 14.62/8.43  | 7.48/0.17   | 7.41/0.02   | 0.02/0.01   |
| BlendedMVS                                             | 9.60/0.38   | 6.65/3.06   | 7.91/0.55   | 36.10/29.55 |
| Olsson                                                 | 13.57/4.37  | 13.23/5.41  | 10.49/2.27  | 12.34/3.35  |
| Registered cameras $N_r$                               |             |             |             |             |
| MegaDepth                                              | 80%         | 79%         | 78%         | 85%         |
| 1DSfM                                                  | 74%         | 77%         | 80%         | 77%         |
| 1DSfM-hard                                             | 38%         | 39%         | 43%         | 37%         |
| Strecha                                                | 91%         | 96%         | 98%         | 100%        |
| BlendedMVS                                             | 95%         | 88%         | 95%         | 90%         |
| Olsson                                                 | 83%         | 90%         | 90%         | 88%         |

Table 13. Training and evaluation configuration. All arms in the main paper use these settings; the only per-method difference is the base learning rate.

| Setting                                | Value                                                                    |
|----------------------------------------|--------------------------------------------------------------------------|
| Training (both methods)                |                                                                          |
| Epochs                                 | 20,000 (one epoch = one pass over all scenes)                            |
| Batch                                  | 4 scenes                                                                 |
| LR schedule                            | milestones 10k, 11k, $\dots$ , 20k, $\gamma = 0.9$                       |
| Base LR                                | $10^{-4}$ (SS-RESfM) / $10^{-3}$ (RESfM)                                 |
| Loss weights                           | $\alpha_{\text{reproj}} = 1, \beta_{\text{cls}} = 1$                     |
| Infinity-point margin                  | $10^{-4}$                                                                |
| Checkpoint selection                   | best validation error, evaluated every 500 epochs                        |
| Seed                                   | 20 (21–22 / 21–24 for the seed studies)                                  |
| Adaptive pseudo-labels (SS-RESfM only) |                                                                          |
| Inlier / outlier percentile            | 20 / 80 (of per-observation reproj. error)                               |
| Threshold separation                   | scale-aware floor, $0.25 \times$ median error                            |
| Warm-up                                | 10 epochs before pseudo-labeling starts                                  |
| Evaluation                             |                                                                          |
| Removal threshold $\tau$               | 0.6 (0.8 for low-contamination single-camera sets, following RESfM)      |
| Test-time fine-tuning                  | 1,001 iterations, LR $5 \times 10^{-3}$ , evaluated every 250 iterations |
| Bundle adjustment                      | pyceres/pycolmap robust BA, as RESfM                                     |

Table 14. Per-scene label-degradation results. For each scene: contamination (fraction of track observations that are GT outliers), COLMAP-label coverage (fraction of observations receiving a label), and precision/recall of the COLMAP-derived outlier labels against GT, sorted by contamination. Above  $\sim 15\%$  contamination, label recall never exceeds 0.50: the labels miss at least half of the true outliers.

| Dataset    | Scene             | Contam. (%) | Coverage | Prec. | Recall |
|------------|-------------------|-------------|----------|-------|--------|
| BlendedMVS | 58c4bb4f4a...     | 0.5         | 1.00     | 0.86  | 0.94   |
| Strecha    | Herz-Jesu-P8      | 0.7         | 1.00     | 0.51  | 0.66   |
| Strecha    | fountain-P11      | 0.7         | 1.00     | 0.69  | 0.76   |
| Strecha    | Herz-Jesu-P25     | 2.6         | 0.99     | 0.72  | 0.82   |
| Strecha    | entry-P10         | 2.7         | 0.99     | 0.88  | 0.88   |
| BlendedMVS | 5be883a4f9...     | 4.0         | 0.98     | 0.90  | 0.84   |
| BlendedMVS | 5b6e716d67...     | 6.1         | 0.92     | 0.89  | 0.92   |
| BlendedMVS | 5bfd0f32ec...     | 7.1         | 0.97     | 0.97  | 0.97   |
| BlendedMVS | 5b950c7160...     | 9.2         | 0.96     | 0.92  | 0.94   |
| MegaDepth  | 0007              | 11.2        | 0.73     | 0.84  | 0.47   |
| 1DSfM      | Alamo             | 26.6        | 0.76     | 0.81  | 0.25   |
| 1DSfM      | Madrid_Metropolis | 28.8        | 0.67     | 0.85  | 0.28   |
| MegaDepth  | 0041              | 31.4        | 0.49     | 0.80  | 0.42   |
| MegaDepth  | 0060              | 39.0        | 0.42     | 0.53  | 0.45   |
| 1DSfM      | Piazza_del_Popolo | 41.9        | 0.76     | 0.90  | 0.30   |
| MegaDepth  | 0047              | 54.9        | 0.37     | 0.83  | 0.50   |
| 1DSfM      | Ellis_Island      | 57.9        | 0.79     | 0.89  | 0.37   |
| 1DSfM-hard | Gendarmenmarkt    | 59.8        | 0.50     | 0.89  | 0.29   |

## J. Per-scene label degradation

Table 14 gives the per-scene data behind the main paper’s label-ceiling analysis: for each of the 18 scenes we measure the contamination level, the fraction of track observations that receive a COLMAP-derived label at all (coverage), and the precision and recall of those labels against ground truth. The aggregate trend in the main paper is visible scene by scene: below  $\sim 10\%$  contamination coverage is 0.92–1.00 and recall is 0.66–0.97 (mostly above 0.8), while above  $\sim 15\%$  contamination recall never exceeds 0.50 on any scene (0.25–0.50) and coverage falls as low as 0.37: the supervision signal degrades exactly where outlier handling matters most.

## K. Extended main-paper tables (camera-ready)

Tables 15–18 extend the main paper’s Tables 2, 5, 6, and 7 with rotation error and registered-camera counts alongside translation, following RESfM’s metric emphasis (translation first, then rotation, then  $N_r$ ): Table 15 extends the main comparison (Table 2), Table 16 the full mechanism matrix (Table 5), Table 17 the removal-statistic ablation (Table 6), and Table 18 the per-scene selection (Table 7). Table 15 reports the seed-20 evaluations behind the main paper; Tables 16 and 17 report mean±std over five independent training seeds (20–24) for every dataset and metric. In the camera-ready we will replace main-paper Tables 2, 5, 6, and 7 with these extended versions, with the translation cells additionally carrying the five-seed training-seed bands of Sec. F (Table 9). Reading the additions (Table 15): rotation broadly agrees with translation (the supervised BlendedMVS failure is even larger in rotation, 36°; the high-contamination ordering is preserved), though the best arm can differ per metric (on MegaDepth the single-draw weight+TTT evaluation leads translation, 0.316 in Table 12, while the supervised baseline leads rotation). Registration is contamination-dependent: the supervised baseline registers the most cameras in distribution (85%) and on clean Strecha (100%), while under high contamination the label-free arms retain more (38–43% vs. 37% on 1DSfM-hard). In the ablation (Table 17) rotation and registration are statistic-insensitive on 1DSfM (band means 7.7–8.9°, 70–75%): the statistic matters through translation. The selection oracle (Table 18) improves both metrics on MegaDepth — translation 0.116 vs. 0.422 for the best fixed arm and 0.344 for the supervised RESfM-from-scratch baseline; rotation 1.22° vs. the baseline’s 2.01° — but registers fewer 1DSfM cameras (66% vs. 74–77% for the fixed arms): selecting by translation error can sacrifice registration.

Two notions of registration, and an error-vs-coverage trade-off. Two registration notions must be distinguished. Main-paper Table 4’s “Registered” column reports pose coverage: every learned arm outputs a pose for every camera (“all”), whereas COLMAP genuinely fails to register 36% of the cameras.  $N_r$  here is stricter: the cameras surviving robust BA and the 5-pixel reprojection quality mask, i.e. the set over which the reported errors are computed. Under this stricter notion (Table 16,  $N_r$  panel) the plain-removal arms attain their low errors over a reduced registered set: on 1DSfM plain remove registers 58% vs. the supervised baseline’s 77%, and on 1DSfM-hard 29% vs. 34% (remove+TTT: 58%/32%), consistent with aggressive pruning disconnecting weakly-supported cameras (main-paper limitation 5). The camera-ready will state this caveat alongside the plain-remove claims (“over the cameras it retains”). madweight, by contrast, registers comparably to the supervised baseline on 1DSfM (70% vs. 77%) and more under extreme contamination (43% vs. 34% on 1DSfM-hard), so its statistical match at extreme contamination and its clean-OOD wins are not coverage artifacts, and the soft arms register at least as many cameras as supervision under high contamination. These readings, together with the paragraph above, are intended for the camera-ready main text (the Metrics paragraph, Secs. 5.1 and 5.5, and limitation 5).

Table 15. Extended main comparison (extends main-paper Table 2; RESfM-aligned schedule, seed 20). Translation panel repeats the main table (mean / median;  $^{\dagger}/^{\ddagger}$  as there); rotation (mean / median,  $^{\circ}$ ) and  $N_r$  panels at seed 20. Best per column in bold.

| Arm                                                   | MegaDepth (25.4)                  | 1DSfM (43.3)      | 1DSfM-hard (60)                                     | Strecha (1.7)      | BMVS (3.1)          | Olsson (0.5)                                       |
|-------------------------------------------------------|-----------------------------------|-------------------|-----------------------------------------------------|--------------------|---------------------|----------------------------------------------------|
| Translation error (mean / median), the primary metric |                                   |                   |                                                     |                    |                     |                                                    |
| madweight                                             | 0.422/0.145                       | 12.39/4.37        | 17.0 $\pm$ 4.4/ <b>11.7<math>^{\ddagger}</math></b> | 2.50/0.97          | 0.146/ <b>0.004</b> | <b>2.8 <math>\pm</math> 0.3</b> /1.2 $^{\ddagger}$ |
| weight                                                | 0.423/0.182                       | 12.79/11.01       | 21.9 $\pm$ 3.4/12.7 $^{\ddagger}$                   | 2.00/0.026         | <b>0.100</b> /0.036 | 2.8 $\pm$ 0.4/1.1 $^{\ddagger}$                    |
| weight+TTT                                            | 0.39 $\pm$ 0.07/0.14 $^{\dagger}$ | 16.13/7.16        | 21.9 $\pm$ 3.9/14.5 $^{\ddagger}$                   | 1.98/ <b>0.005</b> | 0.136/0.011         | 8.0 $\pm$ 0.4/0.4 $^{\ddagger}$                    |
| RESfM-scratch (supervised)                            | <b>0.344/0.070</b>                | <b>11.03/2.93</b> | <b>16.22</b> /13.08                                 | <b>0.006/0.005</b> | 0.379/0.348         | 8.47/ <b>0.17</b>                                  |
| RESfM (released)                                      | 0.209                             | 10.71             | —                                                   | 0.197              | 0.330               | —                                                  |
| RESfM (paper)                                         | 0.169                             | —                 | —                                                   | —                  | —                   | —                                                  |
| Rotation error ( $^{\circ}$ , mean / median)          |                                   |                   |                                                     |                    |                     |                                                    |
| madweight                                             | 3.49/1.33                         | <b>8.31</b> /4.53 | <b>14.40/7.39</b>                                   | 14.62/8.43         | 9.60/0.38           | 13.57/4.37                                         |
| weight                                                | 3.04/1.77                         | 8.39/ <b>2.79</b> | 25.58/17.70                                         | 7.48/0.17          | <b>6.65</b> /3.06   | 13.23/5.41                                         |
| weight+TTT                                            | 2.51/1.39                         | 11.19/11.25       | 31.97/12.83                                         | 7.41/0.02          | 7.91/ <b>0.55</b>   | <b>10.49/2.27</b>                                  |
| RESfM-scratch (supervised)                            | <b>2.01/0.84</b>                  | 9.79/ <b>2.79</b> | 22.70/15.86                                         | <b>0.02/0.01</b>   | 36.10/29.55         | 12.34/3.35                                         |
| Registered cameras $N_r$                              |                                   |                   |                                                     |                    |                     |                                                    |
| madweight                                             | 80%                               | 74%               | 38%                                                 | 91%                | <b>95%</b>          | 83%                                                |
| weight                                                | 79%                               | 77%               | 39%                                                 | 96%                | 88%                 | <b>90%</b>                                         |
| weight+TTT                                            | 78%                               | <b>80%</b>        | <b>43%</b>                                          | 98%                | <b>95%</b>          | <b>90%</b>                                         |
| RESfM-scratch (supervised)                            | <b>85%</b>                        | 77%               | 37%                                                 | <b>100%</b>        | 90%                 | 88%                                                |

Table 16. Extended full shallow mechanism matrix, five-seed bands (extends main-paper Table 5). All entries are mean $\pm$ std over five training seeds (seeds 20–24);  $^{(n)}$  marks cells currently backed by fewer seeds (their remaining seed evaluations are in flight).  $N_r$  is the percentage of the dataset’s ground-truth cameras surviving robust BA and the 5-pixel quality mask.

| Arm                                                    | MD              | 1DSfM            | 1DSfM-hard       | Strecha          | BMVS             | Olsson           |
|--------------------------------------------------------|-----------------|------------------|------------------|------------------|------------------|------------------|
| Translation error (mean $\pm$ std over training seeds) |                 |                  |                  |                  |                  |                  |
| remove                                                 | 0.51 $\pm$ 0.08 | 9.13 $\pm$ 2.17  | 12.42 $\pm$ 2.05 | 3.20 $\pm$ 0.11  | 0.24 $\pm$ 0.06  | 3.41 $\pm$ 0.57  |
| remove+TTT                                             | 0.56 $\pm$ 0.07 | 8.93 $\pm$ 1.20  | 15.15 $\pm$ 0.98 | 3.01 $\pm$ 0.07  | 0.22 $\pm$ 0.10  | 3.63 $\pm$ 0.35  |
| hybrid                                                 | 0.45 $\pm$ 0.08 | 16.23 $\pm$ 2.20 | 22.98 $\pm$ 4.02 | 3.25 $\pm$ 0.45  | 0.14 $\pm$ 0.02  | 2.90 $\pm$ 0.23  |
| hybrid+TTT                                             | 0.43 $\pm$ 0.11 | 15.13 $\pm$ 1.04 | 23.00 $\pm$ 3.06 | 3.13 $\pm$ 0.24  | 0.14 $\pm$ 0.02  | 3.06 $\pm$ 0.39  |
| madweight                                              | 0.51 $\pm$ 0.08 | 11.04 $\pm$ 1.68 | 18.92 $\pm$ 4.05 | 2.65 $\pm$ 0.25  | 0.14 $\pm$ 0.02  | 3.01 $\pm$ 0.34  |
| weight                                                 | 0.51 $\pm$ 0.09 | 14.35 $\pm$ 2.97 | 22.92 $\pm$ 2.79 | 1.97 $\pm$ 0.09  | 0.14 $\pm$ 0.03  | 2.91 $\pm$ 0.33  |
| weight+TTT                                             | 0.40 $\pm$ 0.11 | 15.72 $\pm$ 2.10 | 22.24 $\pm$ 3.17 | 2.00 $\pm$ 0.05  | 0.14 $\pm$ 0.02  | 8.03 $\pm$ 0.30  |
| RESfM (sup.)                                           | 0.37 $\pm$ 0.12 | 10.50 $\pm$ 1.45 | 19.16 $\pm$ 3.42 | 0.39 $\pm$ 0.72  | 0.35 $\pm$ 0.04  | 7.37 $\pm$ 2.63  |
| Rotation error ( $^{\circ}$ , mean $\pm$ std)          |                 |                  |                  |                  |                  |                  |
| remove                                                 | 4.04 $\pm$ 0.60 | 6.94 $\pm$ 1.76  | 12.27 $\pm$ 2.84 | 14.39 $\pm$ 1.03 | 16.04 $\pm$ 4.71 | 13.89 $\pm$ 1.44 |
| remove+TTT                                             | 4.22 $\pm$ 0.36 | 6.90 $\pm$ 1.85  | 16.40 $\pm$ 1.09 | 13.23 $\pm$ 0.78 | 13.19 $\pm$ 6.05 | 14.53 $\pm$ 2.01 |
| hybrid                                                 | 3.08 $\pm$ 0.71 | 7.89 $\pm$ 1.49  | 28.21 $\pm$ 2.10 | 15.15 $\pm$ 1.98 | 9.01 $\pm$ 1.07  | 13.35 $\pm$ 1.60 |
| hybrid+TTT                                             | 2.99 $\pm$ 0.45 | 10.19 $\pm$ 1.10 | 30.63 $\pm$ 3.18 | 14.69 $\pm$ 1.41 | 8.30 $\pm$ 1.21  | 13.73 $\pm$ 2.32 |
| madweight                                              | 3.70 $\pm$ 0.30 | 7.74 $\pm$ 0.94  | 27.65 $\pm$ 8.01 | 12.80 $\pm$ 2.57 | 9.01 $\pm$ 2.37  | 14.44 $\pm$ 0.51 |
| weight                                                 | 3.28 $\pm$ 0.49 | 8.24 $\pm$ 1.97  | 27.95 $\pm$ 3.55 | 7.61 $\pm$ 0.31  | 9.17 $\pm$ 2.22  | 13.24 $\pm$ 0.94 |
| weight+TTT                                             | 2.88 $\pm$ 0.55 | 10.09 $\pm$ 1.30 | 30.60 $\pm$ 3.78 | 7.95 $\pm$ 0.73  | 8.30 $\pm$ 1.21  | 11.60 $\pm$ 1.19 |
| RESfM (sup.)                                           | 1.89 $\pm$ 0.46 | 9.62 $\pm$ 0.63  | 27.30 $\pm$ 5.06 | 1.56 $\pm$ 2.54  | 31.66 $\pm$ 4.19 | 12.62 $\pm$ 0.82 |
| Registered cameras $N_r$ (%), mean $\pm$ std)          |                 |                  |                  |                  |                  |                  |
| remove                                                 | 71.9 $\pm$ 0.9  | 58.0 $\pm$ 1.2   | 28.8 $\pm$ 0.9   | 97.4 $\pm$ 1.7   | 87.8 $\pm$ 2.0   | 78.3 $\pm$ 1.4   |
| remove+TTT                                             | 70.7 $\pm$ 1.8  | 57.5 $\pm$ 2.3   | 32.0 $\pm$ 1.1   | 95.6 $\pm$ 1.7   | 89.7 $\pm$ 2.0   | 78.3 $\pm$ 1.2   |
| hybrid                                                 | 76.1 $\pm$ 1.9  | 80.2 $\pm$ 2.7   | 41.2 $\pm$ 5.0   | 71.5 $\pm$ 3.4   | 93.0 $\pm$ 2.0   | 87.7 $\pm$ 0.7   |
| hybrid+TTT                                             | 76.6 $\pm$ 1.6  | 82.0 $\pm$ 1.2   | 41.5 $\pm$ 4.5   | 70.7 $\pm$ 5.6   | 94.4 $\pm$ 0.8   | 87.7 $\pm$ 1.3   |
| madweight                                              | 81.4 $\pm$ 1.1  | 69.8 $\pm$ 2.4   | 42.6 $\pm$ 4.9   | 91.5 $\pm$ 4.1   | 92.2 $\pm$ 1.7   | 85.7 $\pm$ 1.9   |
| weight                                                 | 78.5 $\pm$ 1.4  | 79.0 $\pm$ 2.9   | 40.2 $\pm$ 2.3   | 97.8 $\pm$ 1.5   | 93.5 $\pm$ 3.3   | 89.9 $\pm$ 1.2   |
| weight+TTT                                             | 76.8 $\pm$ 1.6  | 81.2 $\pm$ 1.3   | 41.2 $\pm$ 3.1   | 98.5 $\pm$ 1.5   | 94.4 $\pm$ 0.9   | 90.0 $\pm$ 0.4   |
| RESfM (sup.)                                           | 84.6 $\pm$ 1.2  | 77.3 $\pm$ 1.6   | 34.3 $\pm$ 3.6   | 99.3 $\pm$ 1.0   | 91.7 $\pm$ 1.8   | 89.3 $\pm$ 1.5   |

Table 17. Extended removal-statistic ablation, five-seed bands (extends main-paper Table 6; head reweighting fixed). Same conventions as Table 16.

| Statistic                                              | MD              | 1DSfM            | 1DSfM-hard       | Strecha          | BMVS             | Olsson           |
|--------------------------------------------------------|-----------------|------------------|------------------|------------------|------------------|------------------|
| Translation error (mean $\pm$ std over training seeds) |                 |                  |                  |                  |                  |                  |
| MAD (madweight)                                        | 0.51 $\pm$ 0.08 | 11.04 $\pm$ 1.68 | 18.92 $\pm$ 4.05 | 2.65 $\pm$ 0.25  | 0.14 $\pm$ 0.02  | 3.01 $\pm$ 0.34  |
| Huber                                                  | 0.59 $\pm$ 0.03 | 14.01 $\pm$ 3.65 | 20.26 $\pm$ 2.87 | 2.71 $\pm$ 0.09  | 0.13 $\pm$ 0.03  | 2.94 $\pm$ 0.51  |
| STD                                                    | 0.53 $\pm$ 0.06 | 13.47 $\pm$ 2.92 | 21.88 $\pm$ 3.06 | 2.42 $\pm$ 0.33  | 0.20 $\pm$ 0.09  | 2.79 $\pm$ 0.45  |
| Rotation error ( $^\circ$ , mean $\pm$ std)            |                 |                  |                  |                  |                  |                  |
| MAD (madweight)                                        | 3.70 $\pm$ 0.30 | 7.74 $\pm$ 0.94  | 27.65 $\pm$ 8.01 | 12.80 $\pm$ 2.57 | 9.01 $\pm$ 2.37  | 14.44 $\pm$ 0.51 |
| Huber                                                  | 4.57 $\pm$ 0.37 | 8.17 $\pm$ 0.92  | 30.66 $\pm$ 4.25 | 14.35 $\pm$ 1.92 | 7.95 $\pm$ 1.54  | 14.13 $\pm$ 1.54 |
| STD                                                    | 3.73 $\pm$ 0.85 | 8.88 $\pm$ 0.81  | 28.27 $\pm$ 3.47 | 10.27 $\pm$ 2.38 | 13.82 $\pm$ 6.80 | 13.05 $\pm$ 1.06 |
| Registered cameras $N_r$ (% , mean $\pm$ std)          |                 |                  |                  |                  |                  |                  |
| MAD (madweight)                                        | 81.4 $\pm$ 1.1  | 69.8 $\pm$ 2.4   | 42.6 $\pm$ 4.9   | 91.5 $\pm$ 4.1   | 92.2 $\pm$ 1.7   | 85.7 $\pm$ 1.9   |
| Huber                                                  | 80.7 $\pm$ 2.1  | 72.9 $\pm$ 2.8   | 38.7 $\pm$ 3.3   | 93.0 $\pm$ 1.5   | 91.5 $\pm$ 5.5   | 85.0 $\pm$ 2.0   |
| STD                                                    | 79.2 $\pm$ 0.8  | 75.4 $\pm$ 1.2   | 39.8 $\pm$ 2.9   | 93.7 $\pm$ 4.6   | 93.0 $\pm$ 2.3   | 87.1 $\pm$ 1.3   |

Table 18. Extended per-scene selection (extends main-paper Table 7; mean translation, then rotation, then  $N_r$ ; seed 20). The oracle (*italics*) selects by ground-truth translation error over five mechanisms; its rotation and  $N_r$  are those of the selected arms.

| Dataset                            | weight       | madwt        | contam     | oracle (GT)  |
|------------------------------------|--------------|--------------|------------|--------------|
| Translation error (means)          |              |              |            |              |
| MegaDepth                          | 0.423        | <b>0.422</b> | 0.422      | <i>0.116</i> |
| 1DSfM                              | 12.79        | <b>12.39</b> | 12.39      | <i>5.78</i>  |
| Strecha                            | <b>2.00</b>  | 2.50         | 2.00       | <i>1.98</i>  |
| BlendedMVS                         | <b>0.100</b> | 0.146        | 0.100      | <i>0.083</i> |
| Rotation error ( $^\circ$ , means) |              |              |            |              |
| MegaDepth                          | <b>3.04</b>  | 3.49         | 3.49       | <i>1.22</i>  |
| 1DSfM                              | 8.39         | <b>8.31</b>  | 8.31       | <i>5.50</i>  |
| Strecha                            | <b>7.48</b>  | 14.62        | 7.48       | <i>7.41</i>  |
| BlendedMVS                         | <b>6.65</b>  | 9.60         | 6.65       | <i>5.28</i>  |
| Registered cameras $N_r$           |              |              |            |              |
| MegaDepth                          | 79%          | <b>80%</b>   | <b>80%</b> | <i>75%</i>   |
| 1DSfM                              | <b>77%</b>   | 74%          | 74%        | <i>66%</i>   |
| Strecha                            | <b>96%</b>   | 91%          | <b>96%</b> | <i>98%</i>   |
| BlendedMVS                         | 88%          | <b>95%</b>   | 88%        | <i>90%</i>   |